

Mathematics Section A

1. Let $\alpha, \alpha + 2, \alpha \in \mathbf{Z}$, be the roots of the quadratic equation $x(x+2) + (x+1)(x+3) + (x+2)(x+4) + \dots + (x+n-1)(x+n+1) = 4n$ for some $n \in \mathbf{N}$. Then $n + \alpha$ is equal to :

(A) 0 (B) 1 (C) 2 (D) 3

2. Let x and y be real numbers such that $50\left(\frac{2x}{1+3i} - \frac{y}{1-2i}\right) = 31 + 17i, i = \sqrt{-1}$. Then the value of $10(x-3y)$ is :

(A) 20 (B) 31 (C) 35 (D) 75

3. Let $\alpha, \beta \in \mathbf{R}$ be such that the system of linear equations $x + 2y + z = 5$

$$2x + y + \alpha z = 5$$

$$8x + 4y + \beta z = 18$$

has no solution. Then $\frac{\beta}{\alpha}$ is equal to :

(A) -4 (B) 4 (C) 8 (D) -8

4. Let $A = \begin{bmatrix} 1 & 2 \\ 1 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 \\ \beta & 2 \end{bmatrix}$. If $A^2 - 4A + I = O$ and $B^2 - 5B - 6I = O$, then among the two statements :

$$(S1) : [(B - A)(B + A)]^T = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$$

and

$$(S2) : \det(\text{adj}(A + B)) = -5,$$

(A) only (S1) is correct (B) only (S2) is correct

(C) both (S1) and (S2) are correct (D) both (S1) and (S2) are wrong

5. Let A be the set of first 101 terms of an A.P., whose first term is 1 and the common difference is 5 and let B be the set of first 71 terms of an A.P., whose first term is 9 and the common difference is 7. Then the number of elements in $A \cap B$, which are divisible by 3, is :

(A) 4 (B) 5 (C) 6 (D) 7

6. The number of seven-digit numbers, that can be formed by using the digits 1, 2, 3, 5 and 7 such that each digit is used at least once, is :

(A) 15400 (B) 17800 (C) 16800 (D) 29400

7. The number of elements in the set $S = \{(r, k) : k \in \mathbf{Z} \text{ and } {}^{36}C_{r+1} = \frac{6({}^{35}C_r)}{(k^2-3)}\}$, is :

(A) 2 (B) 4 (C) 8 (D) 16

8. If the mean of the data

Class	5 - 10	10 - 15	15 - 20	20 - 25	25 - 30	30 - 35
Frequency	2	k	28	54	k + 1	30

is 21, then k is one of the roots of the equation :

(A) $2x^2 - 23x - 10 = 0$ (B) $4x^2 - 35x + 24 = 0$

(C) $2x^2 - 19x - 10 = 0$ (D) $2x^2 - 35x + 98 = 0$

9. Let the mid points of the sides of a triangle ABC $(\frac{5}{2}, 7), (\frac{5}{2}, 3)$ and $(4, 5)$. If its incentre is (h, k) , then $3h$ is equal to :

(A) 11 (B) 12 (C) 13 (D) 14

10. Let an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a < b$, pass through the point $(4, 3)$ and have eccentricity $\frac{\sqrt{5}}{3}$. Then the length of its latus rectum is :

(A) $\frac{4\sqrt{5}}{3}$ (B) $2\sqrt{5}$

(C) $\frac{7\sqrt{5}}{3}$ (D) $\frac{8\sqrt{5}}{3}$

11. If $\sin\left(\frac{\pi}{18}\right)\sin\left(\frac{5\pi}{18}\right)\sin\left(\frac{7\pi}{18}\right) = K$, then the value $\sin\left(\frac{10K\pi}{3}\right)$ is :

(A) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (B) $\frac{\sqrt{3}-1}{\sqrt{2}}$

(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

12. Let $S = \{x \in [-\pi, \pi] : \sin x(\sin x + \cos x) = a, a \in \mathbf{Z}\}$. Then $n(S)$ is equal to :

(A) 3 (B) 6 (C) 7 (D) 9

13. If the point of intersection of the lines $\frac{x+1}{3} = \frac{y+a}{5}$, $\frac{z+b+1}{7}$ and $\frac{x-2}{1} = \frac{y-b}{4} = \frac{z-2a}{7}$ lies on xy -plane, then the value of $a + b$ is :

(A) 2 (B) 5 (C) 7 (D) 9

14. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 2$ and $|\vec{b}| = 3$, then the maximum value of $3|(3\vec{a} + 2\vec{b})| + 4|(3\vec{a} - 2\vec{b})|$ is :

(A) 30 (B) 36 (C) 60 (D) 72

15. Let a line L passing through the point $(1, 1, 1)$ be perpendicular to both the vectors $2\hat{i} + 2\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + 2\hat{k}$. $P(a, b, c)$ is the foot of perpendicular from the origin to the line L , then the value of $34(a + b + c)$ is :

(A) 50 (B) 80 (C) 100 (D) 120

16. If $\lim_{x \rightarrow 2} \frac{\sin(x^3 - 5x^2 + ax + b)}{(\sqrt{x-1}-1)\log_e(x-1)} = m$, then $a + b + m$ is equal to :

(A) 5 (B) 6 (C) 8 (D) 10

17. If the curve $y = f(x)$ passes through the point $(1, e)$ and satisfies the differential equation $\frac{dy}{y(2 + \log_e x)} dx, x > 0$, then $f(e)$ is equal to :

(A) e^e (B) e^{e^2}

(C) e^{2e} (D) e^{e^2}

18. The number of critical points of the function $f(x) = \begin{cases} \left| \frac{\sin x}{x} \right|, & x \neq 0 \\ 1, & x = 0 \end{cases}$ in the interval $(-2\pi, 2\pi)$ is equal to :

- (A) 1 (B) 3 (C) 5 (D) 7

19. Let $[\cdot]$ denote the greatest integer function. Then the value of $\int_0^3 \left(\frac{e^x + e^{-x}}{x!} \right) dx$ is :

- (A) $e^2 + e^3 - \frac{1}{e^2} - \frac{1}{e^3}$
(B) $\frac{1}{2} (e^2 + e^3 - \frac{1}{e^2} - \frac{1}{e^3})$

(C) $e^2 + e^3 - \frac{1}{2e^2} - \frac{1}{2e^3}$

(D) $\frac{1}{2} (e^2 + e^3) - \frac{1}{e^2} - \frac{1}{e^3}$

20. Let $y = y(x)$ be the solution curve of the differential equation $(1 + \sin x) \frac{dy}{dx} + (y + 1) \cos x = 0$, $y(0) = 0$. the curve $y = y(x)$ passes through the point $(\alpha, \frac{-1}{2})$, then a value of α is :

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$
(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Mathematics Section B

21. If the domain of the function $f(x) = \sqrt{\log_{(0.6)} \left(\left| \frac{2x-5}{x^2-4} \right| \right)}$ is $(-\infty, a] \cup \{b\} \cup [c, d) \cup (e, \infty)$, then the value of $a + b + c + d + e$ is _____.

22. If $\sum_{k=1}^n a_k = 6n^3$, then $\sum_{k=1}^6 \left(\frac{a_{k+1} - a_k}{36} \right)^2$ is equal to _____.

23. Let $a, b, c \in \{1, 2, 3, 4\}$. If the probability, that $ax^2 + 2\sqrt{2}bx + c > 0$ for all $x \in \mathbf{R}$, is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to _____.

24. Let a circle C have its centre in the first quadrant, intersect the coordinate axes at exactly three points and cut off equal intercepts from the coordinate axes. If the length of the chord of C on the line $x + y = 1$ is $\sqrt{14}$, then the square of the radius of C is _____.

25. If $\alpha = \int_0^{2\sqrt{3}} \log_2(x^2 + 4) dx + \int_2^4 \sqrt{2x - 4} dx$, then α^2 is equal to _____.

Physics Section A

26. The dimensional formula of $\frac{1}{2}\epsilon_0 E^2$ (ϵ_0 = permittivity of vacuum and E = electric field) is $M^a L^b T^c$. The value of $2a - b + c =$ _____.

- (A) 0 (B) 1 (C) -1 (D) 2

27. The diameter of a wire measured by a screw gauge of least count 0.001 cm is 0.08 cm. The length measured by a scale of least count 0.1 cm is 150 cm. When a weight of 100 N is applied to the wire, the extension in length is 0.5 cm, measured by a micrometer of least count 0.001 cm. The error in the measured Young's modulus is $\alpha \times 10^9$ N/m². The value of α is _____. (Ignore the contribution of the load to Young's modulus error calculation)

- (A) 1.3 (B) 1.65 (C) 0.13 (D) 0.25

28. The velocity of a particle is given as $\vec{v} = -x\hat{i} + 2y\hat{j} - z\hat{k}$ m/s. The magnitude of acceleration at point (1, 2, 4) is _____ m/s².

- (A) $\sqrt{6}$ (B) 9 (C) $\sqrt{33}$ (D) 0

29. The position of an object having mass 0.1 kg as a function of time t is given as $\vec{r} = (10t^2\hat{i} + 5t^3\hat{j})$ m. At $t = 1$ s, which of the following statements are correct?

A. The linear momentum $\vec{p} = (2\hat{i} + 1.5\hat{j})$ kg · m/s. The force acting on the object $\vec{F} = (2\hat{i} + 3\hat{j})$ N. C. The angular momentum of the object about its origin $\vec{L} = 15\hat{k}$ Js. D. The torque acting on the object about its origin $\vec{\tau} = 20\hat{k}$ Nm.

Choose the correct answer from the options given below

- (A) A, B and C only
(B) B, C and D only
(C) A, C and D only
(D) A, B and D only

30. A planet (P_1) is moving around the star of mass $2M$ in the orbit of radius R . Another planet (P_2) is moving around another star of mass $4M$ in a orbit of radius $2R$. Ratio of time periods of revolution of P_2 and P_1 is _____

- (A) $\frac{1}{2}$ (B) 2
(C) 4 (D) $\frac{1}{4}$

31. A particle is rotating in a circular path and at any instant its motion can be described as $\theta = \frac{5t^4}{40} - \frac{t^3}{3}$. The angular acceleration of the particle after 10 seconds is _____ rad/s².

- (A) 150 (B) 120 (C) 130 (D) 170